

LLM Agents in Market Mechanisms vs. Centralized Coordination: A Cross-Paper Analysis

Q Competitive Market Behavior of LLMs [arXiv:2609.02580](#)

P Emission-Forecasting-Based Spatial-Temporal Carbon Response: A Multi-Agent Attention-Enhanced Deep Learning Framework [arXiv:2607.26560](#)

Research phase: PAUSE-ON-ITERATE after 1 round, 1 ITERATE, 0 REVISE

The Pair

Q examines LLM agents in double auction market environments, testing whether such markets deliver efficient allocations when populated by LLM agents rather than humans. The key finding is that markets with LLM agents exhibit slower or no convergence toward market equilibrium, providing less efficient allocations than human-populated markets [\[ledger 1\]](#).

P presents a spatial-temporal carbon response framework using LLM-based multi-agent cooperation for forecasting nodal carbon intensity (NCI). The system coordinates geographically dispatchable loads (mobile energy storage systems and distributed data centers) through centralized optimization, not market mechanisms [\[ledger 2\]](#).

Connexion Pursued

The initial hypothesis proposed testing whether LLM agents coordinating dispatchable loads via market mechanisms (extending P's framework) would exhibit the same convergence failures Q observed in double auctions. This would have feasibility 70 and gain 45, requiring verification of three points: whether P uses market mechanisms, what convergence failures Q documents, and whether a non-obvious result exists beyond incremental extension [\[ledger 1\]](#).

Supported Findings

The ledger establishes a critical structural mismatch:

- Q studies *decentralized market coordination*: agents compete in double auctions, trading with each other under market mechanisms designed for humans.

- P uses *centralized optimization*: LLM agents cooperate to predict NCI, then a central optimizer dispatches loads based on those forecasts.

P does not employ market mechanisms for coordination. The “multi-agent” terminology in P refers to cooperative forecasting agents, not competing traders in a market environment [ledger 2].

Objections

Emmy’s objection (ledger entry 2) identifies a category error in the hypothesis: it conflates “LLM agents” with “market mechanisms.” The presence of LLM agents does not imply market-based coordination. P has the former but not the latter. This objection fundamentally undermines the proposed connexion.

Unresolved Issues

Evidence remains thin on several points:

- Whether P’s centralized LLM coordination exhibits any convergence problems analogous to Q’s market failures (different mechanism, possibly similar dynamics?).
- Whether converting P to a market-based mechanism (if desirable) would inherit Q’s convergence problems.
- Whether the convergence failures in Q stem from model architecture, prompt design, or fundamental misalignment with market incentives.

Smallest Next Step

Re-read P’s methodology section to identify whether any subcomponents use market-like mechanisms (e.g., auction-based load allocation among MESSs/DDCs). If none exist, the hypothesis requires reformulation: either (a) propose a market-based variant of P and predict its behavior using Q’s findings, or (b) investigate whether centralized LLM coordination exhibits different failure modes than decentralized markets.